

INTRO

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This is a simple applet showcasing the power of the [rumtk](#) framework. The goal of this applet is to provide a simple interface for running different kinds of benchmarks on the framework. I also hope to add an export feature to the applet so the results can be saved to disk for record keeping.

This utility is not specific to the [rumtk](#) toolset. You can configure your own profiles to examine your own programs. This benchmarking tool might not be appropriate for interpreted programs since we look at the binary's call stack. As a result, I suspect that the only useful information you might obtain from benchmarking an interpreted program is statistics about the interpreter itself.

THE BENCHMARK REPORT IS BROKEN DOWN INTO THESE SECTIONS:

- Meta: A small section showing the target architecture and OS of this rumtk-benchmark build.
- File List: list of test files and their sizes in MB used for running test of utility.
- Basic: A simple run of the selected utility using hyperfine. The resulting report shows the mean times with breakdowns by user and kernel spaces.
- Flamegraph: Interactive flamegraph attempting to visualize call graph of the test utility under load. Helpful in identifying function targets of interest for future optimizations. This visualization is generated by performing a quick benchmark with the perf command and then passing the report to [flamegraph-rs](#).
- CPU Summary: Simple summary report of basic CPU statistics obtained with the perf command.
- CPU Cache & Branch Misses: Report of functions showing a % cache and branch miss of > 1%. This helps with identifying if the performance issue is coming from a large number of CPU cache misses. Meaning, if the program is not maximizing the usage of the data already present in the CPU cache lines or a failure of the program to pre-train the CPU algorithm to prefetch necessary data. You do get a technical subreport of the current

WISHLIST:

- Network Summary: Report of the local network information.
- Network Stress Test: Test the interfaces against the message and record information about how fast everything moves through the pipes.

INFORMATION THAT IS NOT COLLECTED:

- O Analysis: Currently, we do not gather the information necessary to analyze the general behavior of the program or algorithm. Therefore, this benchmark utility does not provide big O notation information.
- SIMD: Currently, this utility does not inspect and present a disassembly of the test program and in particular the key functions. Thus, no information about proper usage of **SIMD** extensions can be ascertain. I recommend you use Matt Goldbolt's excellent [Compiler Explorer](#) tool or objdump for that.

CONTRIBUTING

If you appreciate the work done in this project, please, visit the following links:

- [OpenCollective](#).
- Navigate to [MedicalMasses L.L.C.'s](#) website.
- Or reach out to [Luis's email](#).

BASIC BENCHMARK

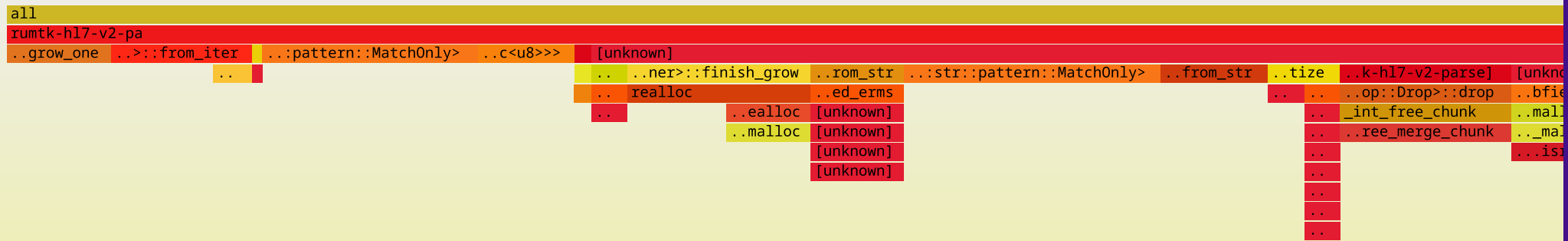
Architecture: x86_64
OS: linux

▼ Test File Sizes in MB

- 2.2236233

Command	Mean	Std	Min	Max	Relative	User	Kernel	Runs	Units
`../target/release/rumtk-hl7-v2-parse`	29.5	1.9	26.4	36.9	1	17.8	11.3	96	[ms]

Flame Graph



```
# started on Tue May  5 10:39:07 2026
```

```
Performance counter stats for '../target/release/rumtk-hl7-v2-parse':
```

```

2,816,607    cache-references:u
 168,746     cache-misses:u
79,922,907   cycles:u
231,579,397  instructions:u
52,300,594   branches:u
   5,499     faults:u
      0      migrations:u

```

```
0.034143674 seconds time elapsed
```

```
0.021979000 seconds user
```

```
0.010978000 seconds sys
```

=====

```
# captured on      : Tue May  5 10:39:08 2026
```

```
# header version : 1
```

```
# data offset      : 680
```

```
# data size      : 9536
```

```
# feat offset      : 10216
```

```
# hostname : fedora
```

```
# os release : 6.19.14-300.fc44.x86_64
```

```
# perf version : 6.19.14-300.fc44.x86_64
```

```
# arch : x86_64
```

```
# nrcpus online : 16
```

```
# nrcpus avail : 16
```

```
# cpudesc : AMD Ryzen 7 7730U with Radeon Graphics
```

```
# cpuid : AuthenticAMD,25,80,0
```

```
# total memory : 39942832 kB
```

```
# cmdline : /usr/bin/perf
```

```
# event : name = cache-misses:u, , id = { 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298 }, type = 4 (cpu), size = 144, config = 0x964 (cache-misses), { sample_period, sample_freq } = 4000, sample_type = IP|TID|TIME|ID|PERIOD, read_format = TOTAL_TIME_ENABLED|TOTAL_TIME_RUNNING|ID|LOST, disabled = 1, inherit = 1, exclude_kernel = 1, exclude_hv = 1, mmap = 1, comm = 1, freq = 1, inherit_stat = 1, enable_on_exec = 1, task = 1, sample_id_all = 1, mmap2 = 1, comm_exec = 1, ksymbol = 1,
```

```
bpf_event = 1, build_id = 1
# event : name = branch-misses:u, , id = { 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314 }, type = 4 (cpu), size = 144, config = 0xc3
(ex_ret_brn_misp), { sample_period, sample_freq } = 4000, sample_type = IP|TID|TIME|ID|PERIOD, read_format = TOTAL_TIME_ENABLED|TOTAL_TIME_RUNNING|ID|LOST, disabled = 1, inherit
= 1, exclude_kernel = 1, exclude_hv = 1, freq = 1, inherit_stat = 1, enable_on_exec = 1, sample_id_all = 1
# sibling sockets : 0-15
# sibling dies : 0-15
# sibling threads : 0-1
# sibling threads : 2-3
# sibling threads : 4-5
# sibling threads : 6-7
# sibling threads : 8-9
# sibling threads : 10-11
# sibling threads : 12-13
# sibling threads : 14-15
# CPU 0: Core ID 0, Die ID 0, Socket ID 0
# CPU 1: Core ID 0, Die ID 0, Socket ID 0
# CPU 2: Core ID 1, Die ID 0, Socket ID 0
# CPU 3: Core ID 1, Die ID 0, Socket ID 0
# CPU 4: Core ID 2, Die ID 0, Socket ID 0
# CPU 5: Core ID 2, Die ID 0, Socket ID 0
# CPU 6: Core ID 3, Die ID 0, Socket ID 0
# CPU 7: Core ID 3, Die ID 0, Socket ID 0
# CPU 8: Core ID 4, Die ID 0, Socket ID 0
# CPU 9: Core ID 4, Die ID 0, Socket ID 0
# CPU 10: Core ID 5, Die ID 0, Socket ID 0
# CPU 11: Core ID 5, Die ID 0, Socket ID 0
# CPU 12: Core ID 6, Die ID 0, Socket ID 0
# CPU 13: Core ID 6, Die ID 0, Socket ID 0
# CPU 14: Core ID 7, Die ID 0, Socket ID 0
# CPU 15: Core ID 7, Die ID 0, Socket ID 0
# node0 meminfo : total = 15444316 kB, free = 310600 kB
# node0 cpu list : 0-15
# node1 meminfo : total = 24498516 kB, free = 13082960 kB
# node1 cpu list : -1
# pmu mappings: cpu = 4, amd_df = 12, amd_iommu_0 = 14, amd_l3 = 13, breakpoint = 5, drm_amdgpu = 4294836224, hwmon_ac = 4294901760, hwmon_acpitz = 4294901761, hwmon_amdgpu =
4294901764, hwmon_bat0 = 4294901762, hwmon_k10temp = 4294901765, hwmon_nvme = 4294901763, hwmon_thinkpad = 4294901766, ibs_fetch = 10, ibs_op = 11, kprobe = 8, msr = 15, power =
16, power_core = 17, software = 1, tool = 4294967294, tracepoint = 2, uprobe = 9
# CPU cache info:
# L1 Data 32K [0-1]
# L1 Instruction 32K [0-1]
# L1 Data 32K [2-3]
# L1 Instruction 32K [2-3]
# L1 Data 32K [4-5]
# L1 Instruction 32K [4-5]
# L1 Data 32K [6-7]
# L1 Instruction 32K [6-7]
# L1 Data 32K [8-9]
# L1 Instruction 32K [8-9]
# L1 Data 32K [10-11]
# L1 Instruction 32K [10-11]
# L1 Data 32K [12-13]
# L1 Instruction 32K [12-13]
# L1 Data 32K [14-15]
# L1 Instruction 32K [14-15]
# L2 Unified 512K [0-1]
# L2 Unified 512K [2-3]
# L2 Unified 512K [4-5]
# L2 Unified 512K [6-7]
# L2 Unified 512K [8-9]
# L2 Unified 512K [10-11]
```

```

# L2 Unified          512K [12-13]
# L2 Unified          512K [14-15]
# L3 Unified          16384K [0-15]
# time of first sample : 68376.152303
# time of last sample  : 68376.180821
# sample duration :    28.518 ms
# memory nodes (nr 2, block size 0x8000000):
#   0 [15G]: 198-317
#   1 [24G]: 0-25,32-197
# bpf_prog_info empty
# btf info empty
# cpu pmu capabilities: max_precise=0
# AMD systems uses ibs_op// PMU for some precise events, e.g.: cycles:p, see the 'perf list' man page for further details.
# missing features: TRACING_DATA BUILD_ID BRANCH_STACK GROUP_DESC AUXTRACE STAT CLOCKID DIR_FORMAT COMPRESSED CLOCK_DATA HYBRID_TOPOLOGY
# =====
#
#
# Total Lost Samples: 0
#
# Samples: 78 of event 'cache-misses:u'
# Event count (approx.): 131531
#
# Overhead  Command          Shared Object          Symbol
# .....
#
#
13.65% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x14b973      B [.] __memmove_avx_unaligned_erms
 8.89% rumtk-hl7-v2-pa [unknown]                    0xffffffff86a01280 ! [k] 0xffffffff86a01280
 8.45% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x8033e      B [.] _int_free_create_chunk
 6.10% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x9831f      B [.] <core::str::pattern::TwoWaySearcher>::next:::
<core::str::pattern::MatchOnly>
 6.04% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x963eb      B [.]
<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Message>::sanitize
 5.25% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x8027b      B [.] unlink_chunk.isra.0
 5.11% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x133e4d      B [.] core::slice::memchr::memchr
 4.74% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x82647      B [.] _int_malloc
 4.65% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x91358      B [.]
<alloc::vec::Vec<alloc::vec::Vec<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field>> as core::ops::drop::Drop>::drop
 3.71% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x814c5      B [.] _int_free_merge_chunk
 3.57% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x91720      B [.] __rustc::__rust_dealloc
 3.10% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x84340      B [.] serde_json::ser::format_escaped_str::<&mut
alloc::vec::Vec<u8>, serde_json::ser::CompactFormatter>
 2.95% rumtk-hl7-v2-pa /usr/lib64/ld-linux-x86-64.so.2    0x5399      B [.] call_init
 2.87% rumtk-hl7-v2-pa [unknown]                    0xffffffff86da6a3e ! [k] 0xffffffff86da6a3e
 2.14% rumtk-hl7-v2-pa [unknown]                    0xffffffff86a00005 ! [k] 0xffffffff86a00005
 2.02% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x9996d      B [.] <core::str::iter::Split<char> as
core::iter::traits::iterator::Iterator>::next
 2.00% rumtk-hl7-v2-pa [unknown]                    0xffffffff86a01284 ! [k] 0xffffffff86a01284
 1.89% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x96235      B [.]
<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Message>::tokenize_segments
 1.84% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x6f67f      B [.] _dlderror_run
 1.68% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x81613      B [.] _int_free_chunk
 1.67% rumtk-hl7-v2-pa [unknown]                    0xffffffff86b04104 ! [k] 0xffffffff86b04104
 1.45% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x99416      B [.] <alloc::vec::Vec<&str> as
alloc::vec::spec_from_iter_nested::SpecFromIterNested<&str, core::str::iter::Split<&str>>::from_iter
 1.41% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x85e98      B [.] <serde_json::ser::Compound<&mut
alloc::vec::Vec<u8>, serde_json::ser::CompactFormatter> as serde_core::ser::SerializeMap>::serialize_entry::<str, compact_str::CompactString>

```

```

# Samples: 79 of event 'branch-misses:u'
# Event count (approx.): 78996
#
# Overhead  Command                Shared Object                                Symbol
# .....
# .....
#
17.52% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x9947f      B [.] <alloc::vec::Vec<&str> as
alloc::vec::spec_from_iter_nested::SpecFromIterNested<&str, core::str::iter::Split<&str>>::from_iter
12.84% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x98380      B [.] <core::str::pattern::TwoWaySearcher>::next::
<core::str::pattern::MatchOnly>
9.51% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x842ee      B [.] serde_json::ser::format_escaped_str::<&mut
alloc::vec::Vec<u8>, serde_json::ser::CompactFormatter>
5.36% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x14b144     B [.] __memcmp_avx2_movbe
4.97% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x83eeb      B [.] malloc
4.21% rumtk-hl7-v2-pa [unknown]          0xffffffff86a01280 ! [k] 0xffffffff86a01280
4.02% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x85ba4      B [.] <serde_json::ser::Compound<&mut
alloc::vec::Vec<u8>, serde_json::ser::CompactFormatter> as serde_core::ser::SerializeMap>::serialize_entry::<str,
alloc::vec::Vec<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Component>>
3.78% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x1325f4     B [.] <core::str::pattern::StrSearcher>::new
3.28% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x133d6a     B [.] core::slice::memchr::memchr_aligned
3.27% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x99fc3      B [.] <core::str::pattern::StrSearcher as
core::str::pattern::Searcher>::next_match
3.23% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x829ef      B [.] _int_malloc
2.51% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x814c5      B [.] _int_free_merge_chunk
2.38% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x858cc      B [.] <serde_json::ser::Compound<&mut
alloc::vec::Vec<u8>, serde_json::ser::CompactFormatter> as serde_core::ser::SerializeMap>::serialize_entry::<str,
alloc::vec::Vec<alloc::vec::Vec<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field>>>
2.23% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x9c4df      B [.] <once_cell::race::once_box::OnceBox<[[u64; 4];
2]>>::init::<<once_cell::race::once_box::OnceBox<_>>::get_or_init::Void, <once_cell::race::once_box::OnceBox<[[u64; 4]; 2]>>::get_or_init<ahash::random_state::get_fixed_seeds::
{closure#0}>>::<closure#0>>
2.18% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x9d37d      B [.] rumtk_core::cli::cli_utils::read_some_stdin
2.12% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x98233      B [.] <phf::map::Map<&str, u8>>::get_entry::<str>
1.86% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x963bb      B [.]
<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Message>::sanitize
1.61% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x876e1      B [.]
<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Component as serde_core::ser::Serialize>::serialize::<&mut serde_json::ser::Serializer<&mut alloc::vec::Vec<u8>>>
1.51% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x94182      B [.]
<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field>::from_str
1.34% rumtk-hl7-v2-pa [unknown]          0xffffffff8726de63 ! [k] 0xffffffff8726de63
1.30% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x912ba      B [.]
<alloc::vec::Vec<alloc::vec::Vec<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field>> as core::ops::drop::Drop>::drop
1.28% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x786eb      B [.] __do_global_dtors_aux
1.24% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x94503      B [.]
<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Segment>::generate_subfields
1.21% rumtk-hl7-v2-pa [unknown]          0xffffffff87165a56 ! [k] 0xffffffff87165a56
1.09% rumtk-hl7-v2-pa /usr/lib64/libc.so.6          0x8033e      B [.] _int_free_create_chunk
1.03% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x86e88      B [.] <rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field
as serde_core::ser::Serialize>::serialize::<&mut serde_json::ser::Serializer<&mut alloc::vec::Vec<u8>>>
1.00% rumtk-hl7-v2-pa [unknown]          0xffffffff880e5b1f ! [k] 0xffffffff880e5b1f
#
# (Tip: If you prefer Intel style assembly, try: perf annotate -M intel)
#

```

